

Intermediate Algebra Test III

Name
Date
Course

I) Write your name, the date, and the course title (i.e. Intermediate Algebra).

II) Show all work.

III) If a problem requires steps to solve, draw a rectangle around your final answer to distinguish it from the other work.

IV) You will need a calculator for this test.

V) You do not have to write the Roman numeral for each exponent rule that you use on this test.

VI) Do not leave your answers with fraction exponents, negative exponents, zero power exponents, or 1st power exponents.

VII) Each problem is worth four points for a total of 100 points.

Simplify.

① $(-64)^{\frac{2}{3}}$

② $\left(\frac{a^2}{b^3}\right)^{-4}$

③ $\sqrt[3]{125 \cdot 125}$

④ $7^{\frac{2}{3}} \cdot 7^{\frac{1}{9}}$

* Remember,
write in
root form.

⑤ $\left(\frac{x^5}{4^{-2}}\right)^1$

⑥ $\frac{2x^2}{y^{-2}}$

⑦ $\left(\frac{2^0 x^{-8}}{y}\right)^3$

⑧ Find the value of the following logarithm without a calculator.
 $\log_6 6^8$

⑨ Combine the expression below into one logarithm.
 $-\log_9 X + \log_9 2$

⑩ If $\log_k 5 \approx 1.46$ and $\log_k 4 \approx 1.26$, approximate the value of the logarithm below.

$$\log_k \left(\frac{5}{4} \right)$$

Solve.

$$\textcircled{11} 16 \cdot 4^{2x-2} = 4^{5x+1}$$

$$\textcircled{12} \frac{1}{3^{x-2}} = (\sqrt[3]{9})^x$$

$$\textcircled{13} \frac{x}{4} = \log_6 \sqrt[4]{6^7}$$

$$\textcircled{14} 81^{2x+1} = \sqrt{27}$$

Write your answer as
an exact expression.

⑮ $\frac{3^{4x^2}}{3^{x+5}} = \frac{1}{81}$

For problem #'s 16-17, round your answers to the
nearest ten thousandth.

⑯ $0 = 9^{2x} + 9^x - 20$

⑰ $4 = -\frac{5^x}{2} + 1$

For problem #'s 18-20, write your answers as exact expressions.

$$(18) 2\log_4(x+1) + \log_4 10 = \log_4 100$$

$$(19) \frac{\log_x 2}{3} - 4 = -3$$

$$(20) -5\log_4(3x-6) + 2 = 0$$

$$\textcircled{21} \quad 1 = \log_6(x^2 + 7x + 16) \quad \textcircled{22} \quad 2 = \log_3 x + 4 \log_3 2$$

$$\textcircled{23} \quad \log_5 x - \log_5(x+4) = 1$$

Simplify.

$$\textcircled{24} \left(\frac{4x^2}{y^2} \right)^{-3} \cdot \left(\frac{x^{-1}}{2b^0} \right)^2 (-b^{-3}xy)$$

$$\textcircled{25} (2a^{\frac{1}{2}} \cdot a^0 \cdot b^4)^2 (3b \cdot a^7 \cdot b^3)^3$$

Answers

① 16

② $\frac{b^{12}}{a^8}$

③ 25

④ $\sqrt[9]{7^7}$ or $(\sqrt[9]{7})^7$

⑤ $16x^5$

⑥ $2x^2y^2$

⑦ $\frac{1}{y^3x^{24}}$

⑧ 8

⑨ $\log_9\left(\frac{2}{x}\right)$

⑩ 0.2

⑪ $x = -\frac{1}{3}$

⑫ $x = \frac{6}{5}$

⑬ $x = 7$

⑭ $x = -\frac{5}{16}$

⑮ $x = \frac{1 \pm \sqrt{17}}{8}$

⑯ $x \approx 0.6309$

⑰ $x \approx 1.1133$

⑱ $x = -1 + \sqrt{10}$

⑲ $x = \sqrt[3]{2}$

⑳ $x = \frac{\sqrt[5]{16} + 6}{3}$ or $\frac{(\sqrt[5]{4})^2 + 6}{3}$

㉑ $x = -2$ or $x = -5$

㉒ $x = \frac{9}{16}$

㉓ No Solution, $x \neq -5$

㉔ $-\frac{y^7}{256x^7b^3}$

㉕ $108b^{20}\sqrt[3]{a^{65}}$ or $108b^{20}(\sqrt[3]{a})^{65}$